NETWORK INFRASTRUCTURE / TECHNICAL PROJECT

Cisco ISR 1100

Secure Data Sanitization & IOS XE Firmware Restoration

CISCO C1111-8PWB

IOS XE 16.09.05 / CONSOLE AND ROMMON MAINTENANCE

A documented enterprise-router maintenance workflow covering boot recovery, firmware handling, cryptographic key maintenance, configuration removal and infrastructure verification.

PHYSICAL DEVICE / FRONT AND REAR VIEWS



FRONT VIEW



REAR INTERFACES

DOCUMENTATION STANDARD

19 redacted terminal captures are presented in sequence. The report distinguishes recorded actions from outcomes that are not independently confirmed.

PROJECT PROFILE

Scope and technical record

This report documents a Cisco ISR 1100 maintenance sequence recorded through serial-console and ROMMON access. The material is organized as a technical project record rather than a device-agnostic reset procedure.

PLATFORM	Cisco ISR 1100 / C1111-8PWB
SOFTWARE	Cisco IOS XE 16.09.05
ACCESS	Serial console, privileged EXEC and ROMMON
STORAGE	Internal bootflash and removable USB

MAINTENANCE WORKSTREAMS

01 Recovery and discovery
ROMMON, software identification, modules and interfaces

02 Firmware and storage
Internal flash inventory, USB image backup and return

03 Configuration maintenance
Selected key, NVRAM, startup-config and bootflash operations

04 Post-operation review
Filesystem, VLAN, module and ROMMON inspection

EVIDENCE QUALIFICATION

Captured operations include configuration and storage maintenance. Transfer progress is visible, but the images do not independently verify completed file copies, image integrity, a final successful IOS XE boot or separate wireless-module sanitization.

NAVIGATION

Contents

Select any step to jump directly to its command explanation and terminal evidence.

01 ROMMON configuration register

02 IOS XE version verification

03 Hardware inventory

04 Platform and wireless module inspection

05 Interface status

06 Ethernet diagnostics

07 IPv4 routing table

08 Running configuration

09 Filesystem inventory

10 IOS XE image backup to USB

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13 NVRAM and startup-config removal

14 Bootflash formatting

15 IOS XE image restoration

16 Post-operation filesystem review

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18 Module review and reload

19 ROMMON environment review

REPORT CLOSEOUT Results, limitations and source record

STEP 01 / MAINTENANCE PROCEDURE

ROMMON configuration register

RECORDED CISCO CLI

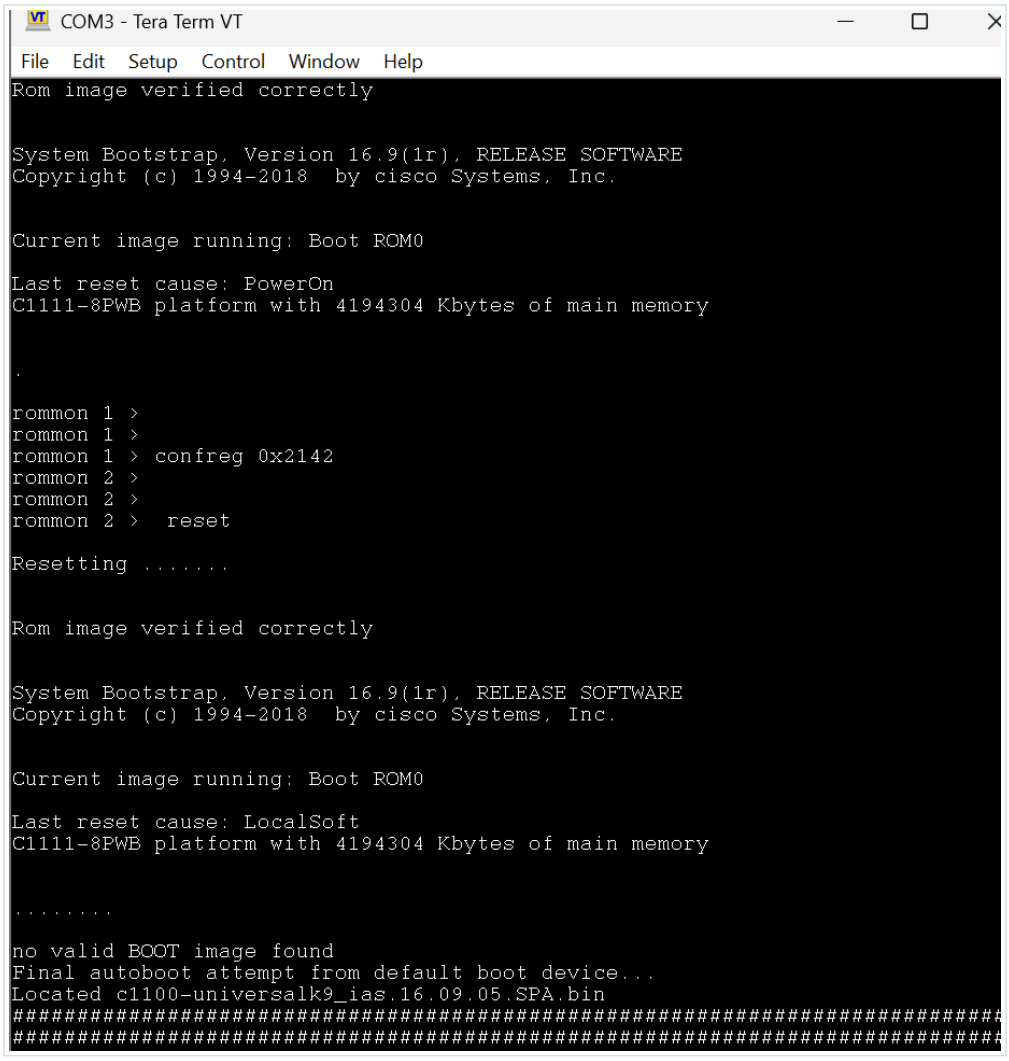
```
rommon 1 > confreg 0x2142
rommon 2 > reset
```

WHY THESE COMMANDS WERE USED

confreg 0x2142 - Bypassed the saved startup configuration during recovery access.

reset - Restarted the router so the recovery register would take effect.

CONSOLE EVIDENCE / REDACTED CAPTURE



STEP 02 / MAINTENANCE PROCEDURE

IOS XE version verification

RECORDED CISCO CLI

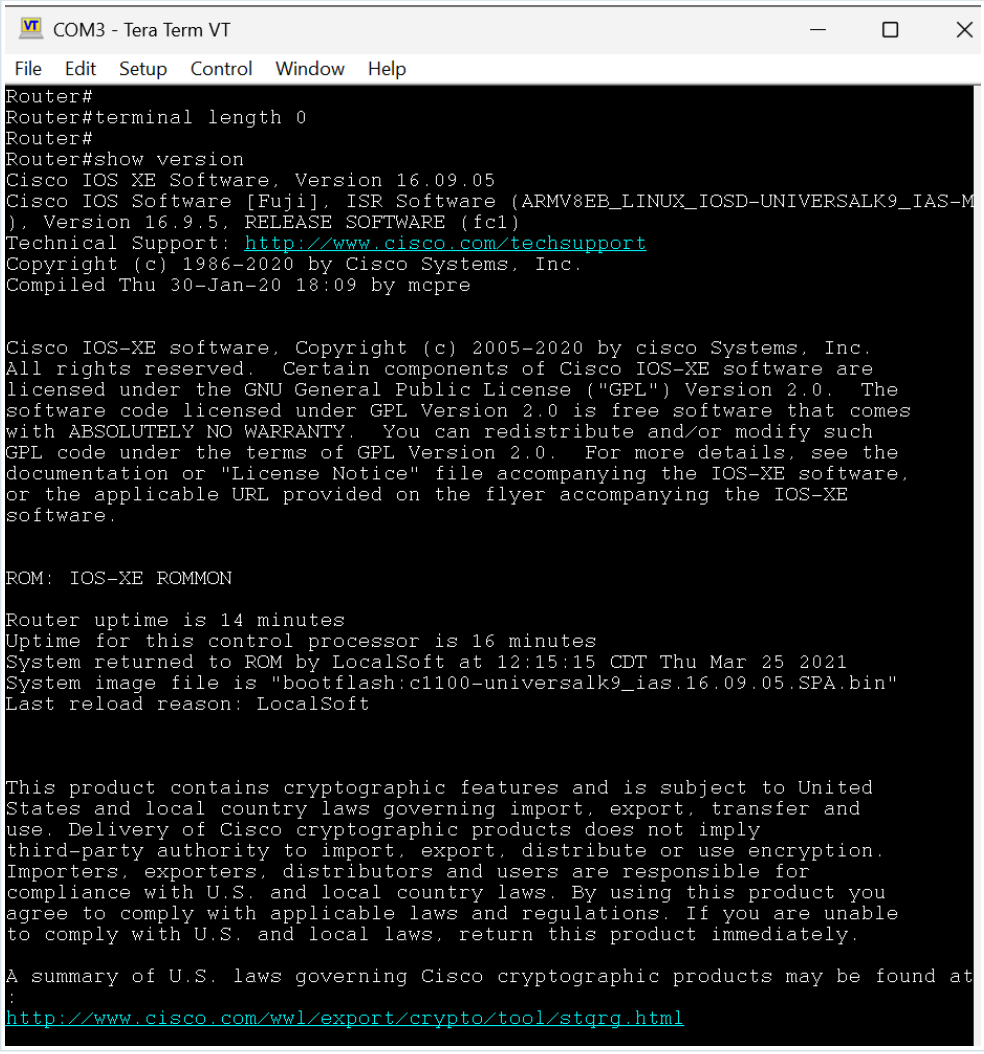
```
Router# terminal length 0
Router# show version
```

WHY THESE COMMANDS WERE USED

terminal length 0 - Removed output pagination for an uninterrupted console capture.

show version - Identified the chassis, IOS XE version and boot details.

CONSOLE EVIDENCE / REDACTED CAPTURE



STEP 03 / MAINTENANCE PROCEDURE

Hardware inventory

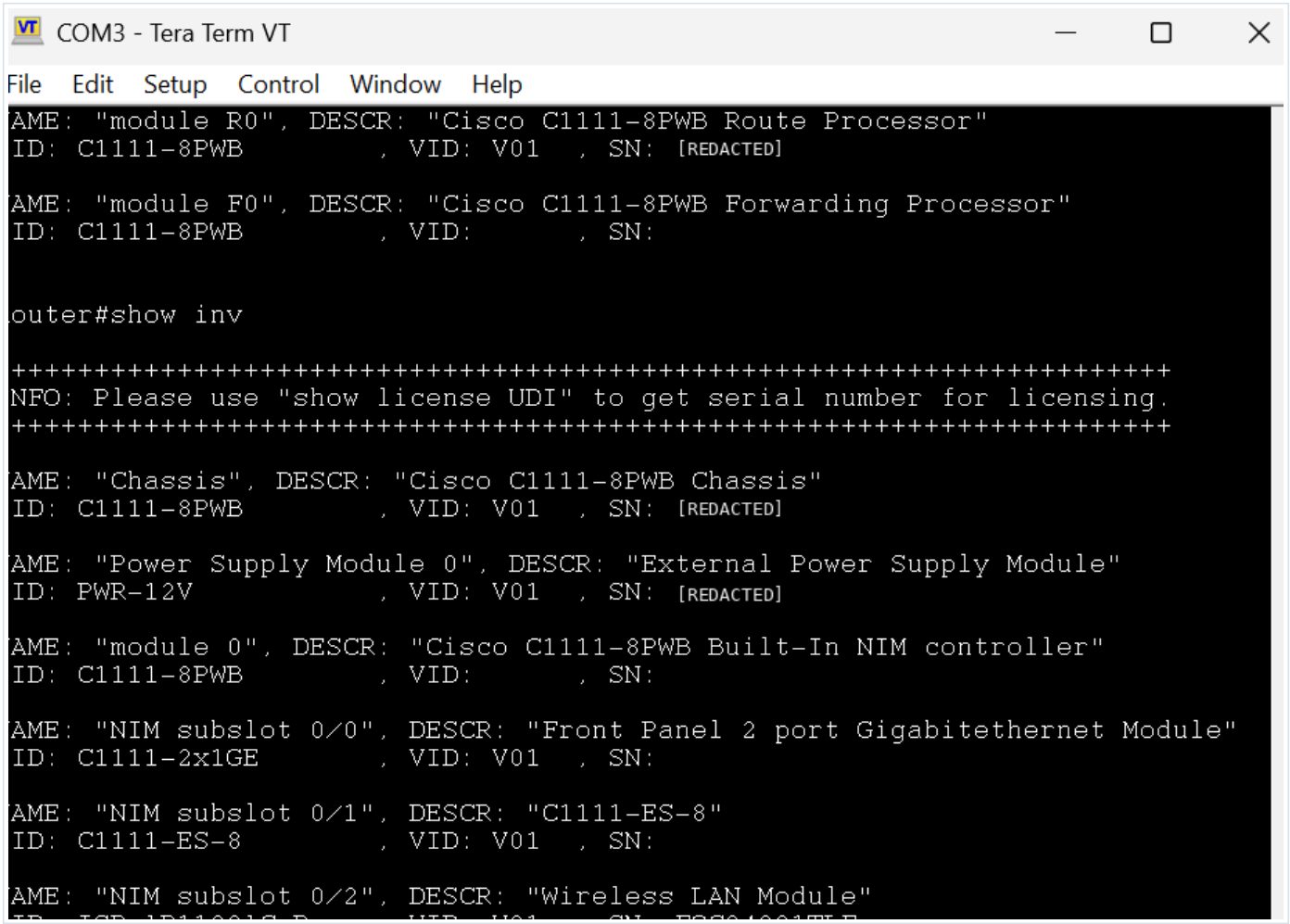
RECORDED CISCO CLI

```
Router# show inventory
```

WHY THESE COMMANDS WERE USED

show inventory - Enumerated the chassis and installed components; unique identifiers were redacted.

CONSOLE EVIDENCE / REDACTED CAPTURE



STEP 04 / MAINTENANCE PROCEDURE

Platform and wireless module inspection

RECORDED CISCO CLI

```
Router# show platform
```

WHY THESE COMMANDS WERE USED

show platform - Checked module presence and operating states, including the embedded wireless component.

CONSOLE EVIDENCE / REDACTED CAPTURE

```
Router#show platform
Chassis type: C1111-8PWB

Slot      Type                State                 Insert time (ago)
--------- ------------------- --------------------- -----------------
0         C1111-8PWB          ok                    00:17:40
 0/0      C1111-2x1GE         ok                    00:16:10
 0/1      C1111-ES-8          ok                    00:16:10
 0/2      ISR-AP1100AC-B      ok                    00:16:10
R0        C1111-8PWB          ok, active            00:17:40
F0        C1111-8PWB          ok, active            00:17:40
P0        PWR-12V             ok                    00:17:20

Slot      CPLD Version        Firmware Version
--------- ------------------- ----------------------------------------
0         18092702            16.9(1r)
R0        18092702            16.9(1r)
F0        18092702            16.9(1r)

Router#
Router#
Router#
Router#
Router#
```

STEP 05 / MAINTENANCE PROCEDURE

Interface status

RECORDED CISCO CLI

```
Router# show ip interface brief
```

WHY THESE COMMANDS WERE USED

show ip interface brief - Established the baseline IP addresses and interface / protocol states.

CONSOLE EVIDENCE / REDACTED CAPTURE

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Proto
col
GigabitEthernet0/0/0   unassigned      YES unset  down                  down

GigabitEthernet0/0/1   unassigned      YES unset  down                  down

GigabitEthernet0/1/0   unassigned      YES unset  down                  down

GigabitEthernet0/1/1   unassigned      YES unset  down                  down

GigabitEthernet0/1/2   unassigned      YES unset  down                  down

GigabitEthernet0/1/3   unassigned      YES unset  down                  down

GigabitEthernet0/1/4   unassigned      YES unset  down                  down

GigabitEthernet0/1/5   unassigned      YES unset  down                  down

GigabitEthernet0/1/6   unassigned      YES unset  down                  down

GigabitEthernet0/1/7   unassigned      YES unset  down                  down

Wl0/1/8                unassigned      YES unset  up                    up

Vlan1                  unassigned      YES unset  up                    up

Router#
```

STEP 06 / MAINTENANCE PROCEDURE

Ethernet diagnostics

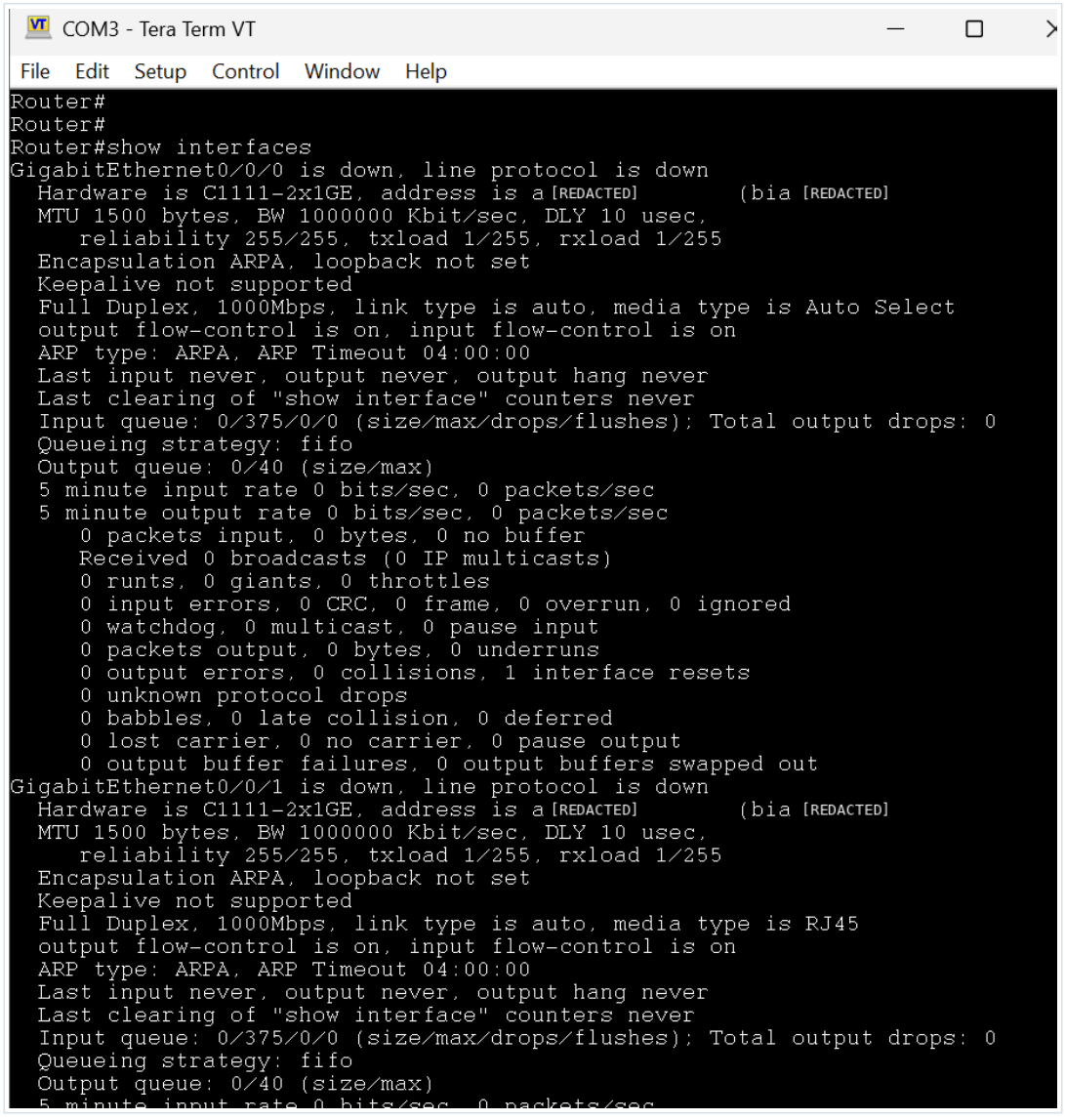
RECORDED CISCO CLI

```
Router# show interfaces
```

WHY THESE COMMANDS WERE USED

show interfaces - Reviewed Ethernet link conditions, counters and error statistics.

CONSOLE EVIDENCE / REDACTED CAPTURE



STEP 07 / MAINTENANCE PROCEDURE

IPv4 routing table

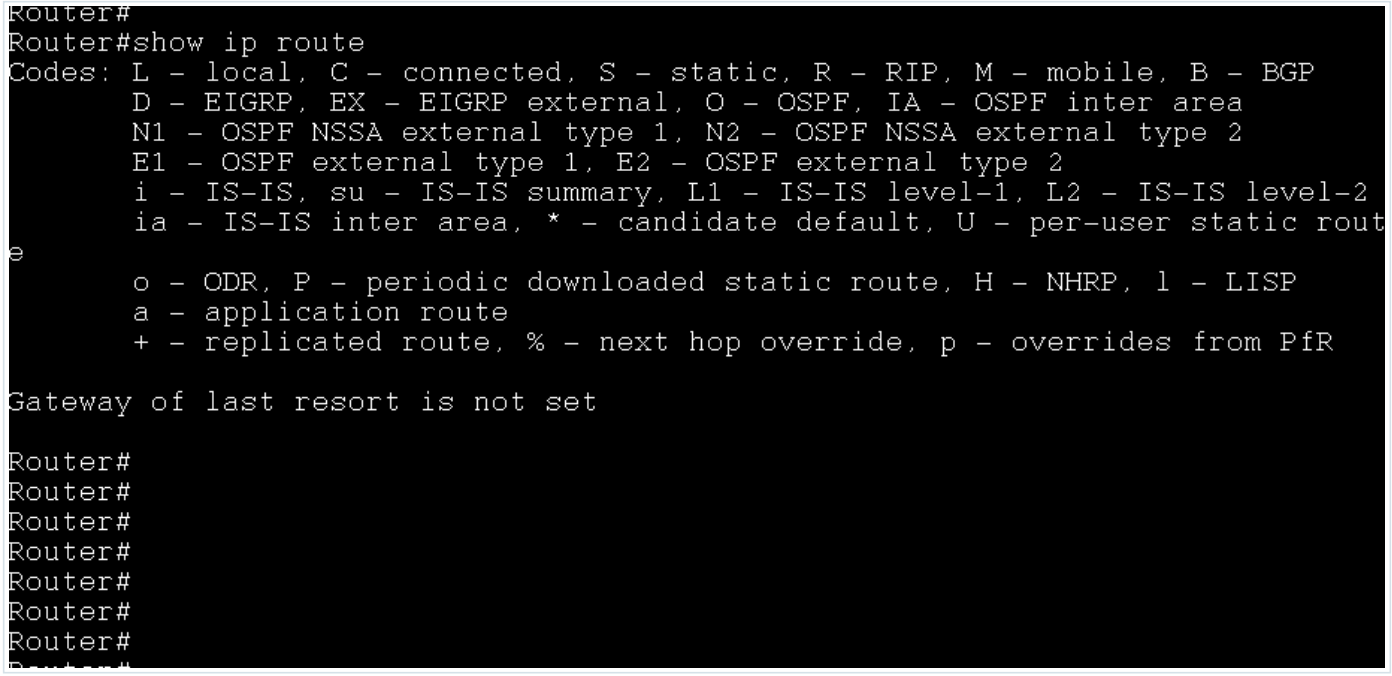
RECORDED CISCO CLI

```
Router# show ip route
```

WHY THESE COMMANDS WERE USED

show ip route - Examined IPv4 routes and default-route status before configuration changes.

CONSOLE EVIDENCE / REDACTED CAPTURE



STEP 08 / MAINTENANCE PROCEDURE

Running configuration

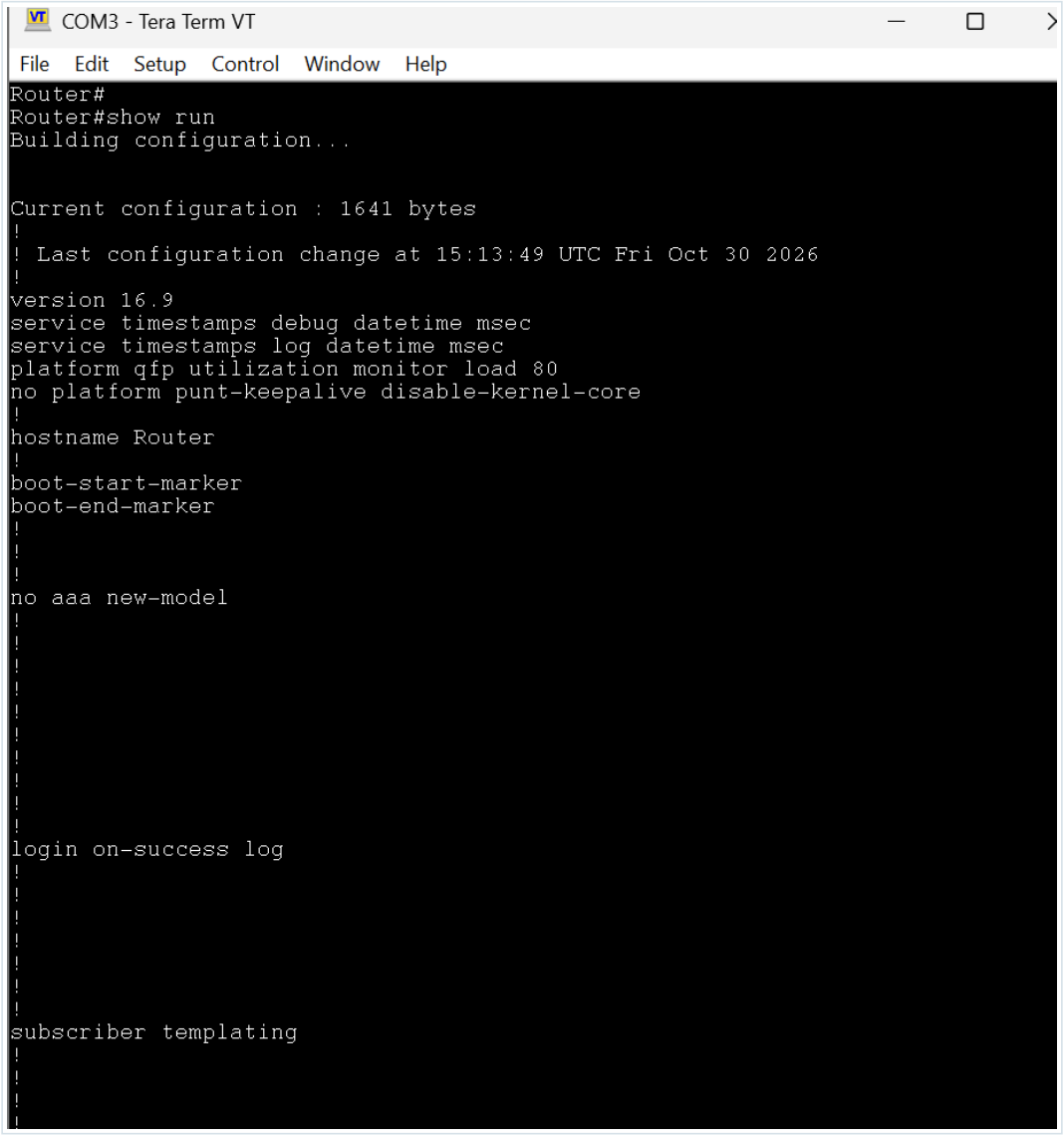
RECORDED CISCO CLI

```
Router# show running-config
```

WHY THESE COMMANDS WERE USED

show running-config - Inspected active IOS XE settings separately from the saved startup configuration.

CONSOLE EVIDENCE / REDACTED CAPTURE



STEP 09 / MAINTENANCE PROCEDURE

Filesystem inventory

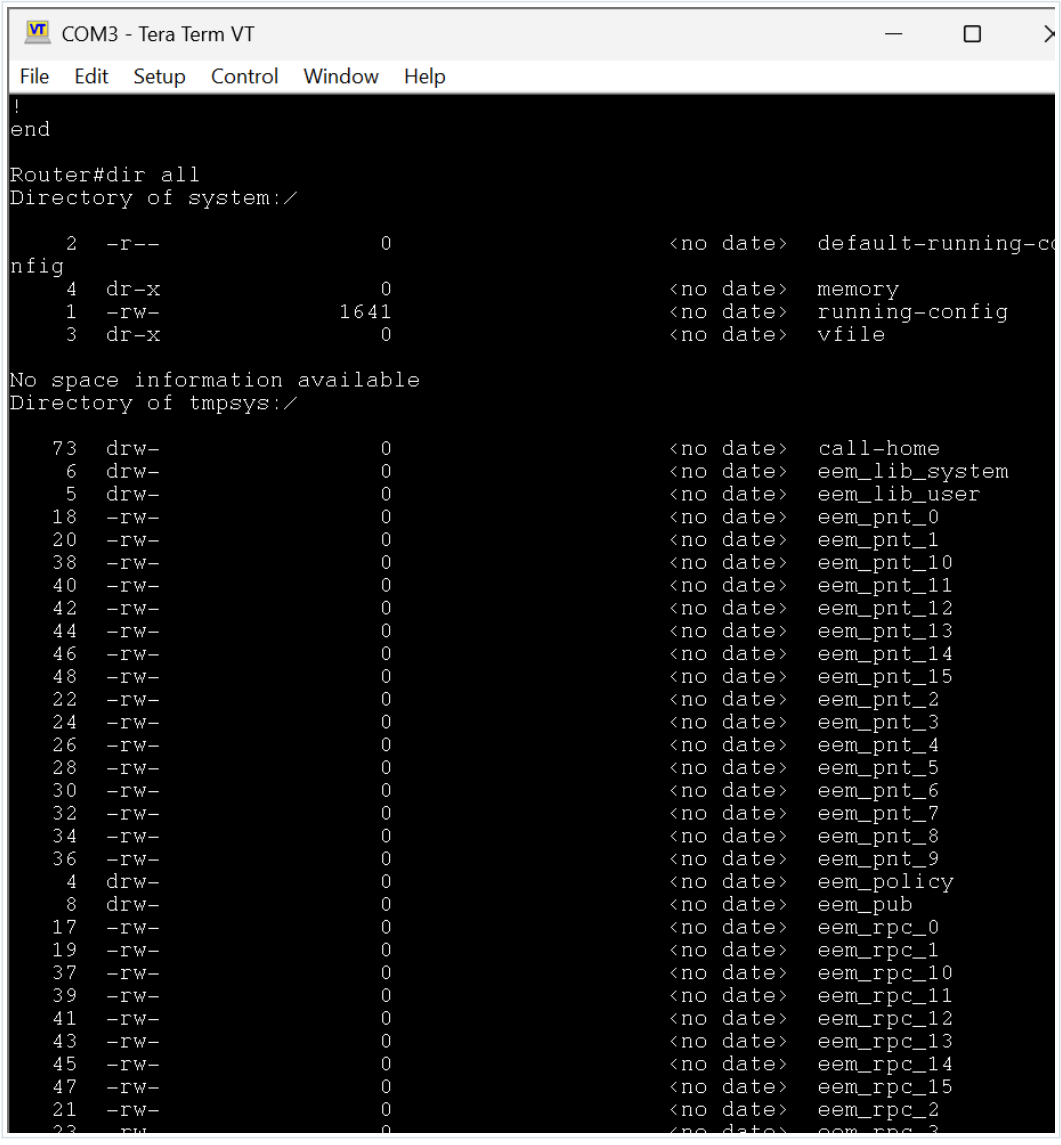
RECORDED CISCO CLI

```
Router# dir all
```

WHY THESE COMMANDS WERE USED

dir all - Enumerated accessible filesystem entries before image transfer and storage maintenance.

CONSOLE EVIDENCE / REDACTED CAPTURE



STEP 10 / MAINTENANCE PROCEDURE

IOS XE image backup to USB

RECORDED CISCO CLI

```
Router# dir bootflash:
Router# copy bootflash: usb0:
Source filename []? c1100-universalk9_ias.16.09.05.SPA.bin
Destination filename [c1100-universalk9_ias.16.09.05.SPA.bin]?
```

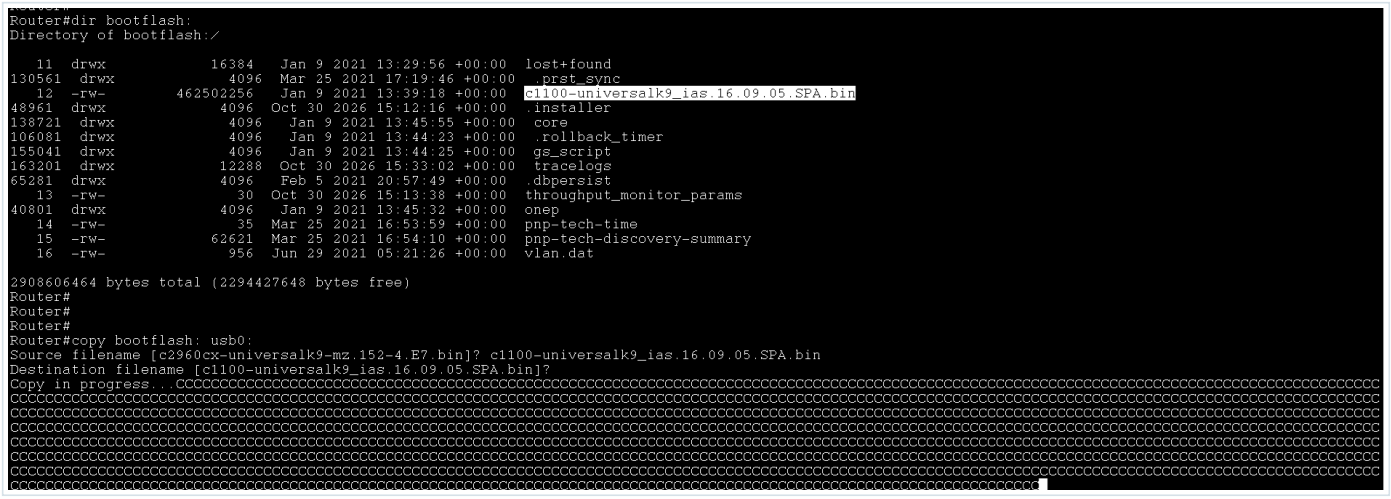
WHY THESE COMMANDS WERE USED

dir bootflash: - Located the IOS XE image stored on internal flash.

copy bootflash: usb0: - Started transferring the selected image to removable USB storage.

Filename prompts - Specified the source .bin image and accepted the destination filename.

CONSOLE EVIDENCE / REDACTED CAPTURE



STEP 11 / MAINTENANCE PROCEDURE

Cryptographic key maintenance

RECORDED CISCO CLI

```
Router# configure terminal
Router(config)# crypto key zeroize ?
Router(config)# crypto key zeroize ec
Router(config)# crypto key zeroize pubkey-chain
Router(config)# crypto key zeroize rsa
```

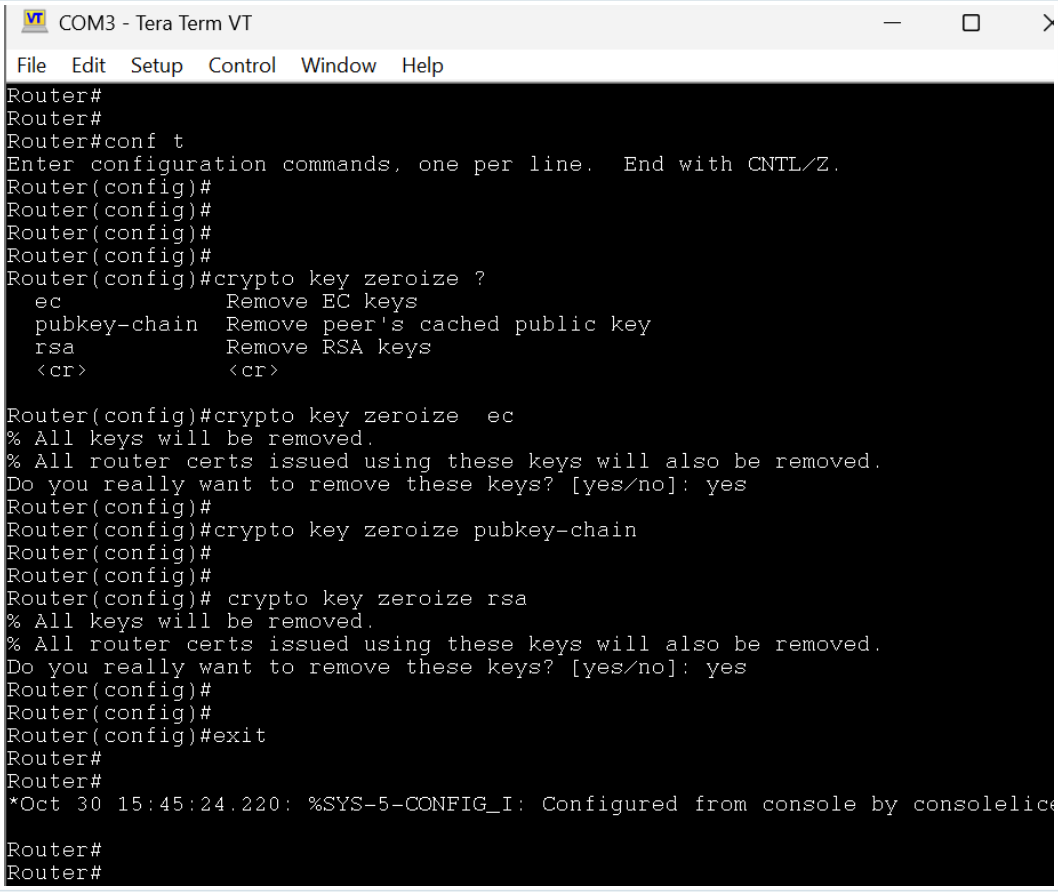
WHY THESE COMMANDS WERE USED

configure terminal - Entered global configuration mode.

crypto key zeroize ? - Listed supported key categories on the device.

crypto key zeroize ec / pubkey-chain / rsa - Addressed the recorded EC, public-key-chain and RSA material.

CONSOLE EVIDENCE / REDACTED CAPTURE



STEP 12 / MAINTENANCE PROCEDURE

VTP and boot-register configuration

RECORDED CISCO CLI

```
Router# configure terminal
Router(config)# vtp domain null
Router(config)# config-register 0x2102
Router(config)# end
```

WHY THESE COMMANDS WERE USED

configure terminal - Entered global configuration mode.

vtp domain null - Set the VTP domain value shown in the captured session.

config-register 0x2102 - Restored the normal configuration-register boot setting.

end - Returned to privileged EXEC mode.

CONSOLE EVIDENCE / REDACTED CAPTURE

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#
Router(config)#vtp domain null
Changing VTP domain name from [REDACTED]       to null
Router(config)#
*Oct 30 15:46:29.982: %SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed
 to null.
Router(config)#
Router(config)#
Router(config)#config-register 0x2102
Router(config)#
Router(config)#
Router(config)#
Router(config)# end
Router#
*Oct 30 15:46:44.970: %SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
```

STEP 13 / MAINTENANCE PROCEDURE

NVRAM and startup-config removal

RECORDED CISCO CLI

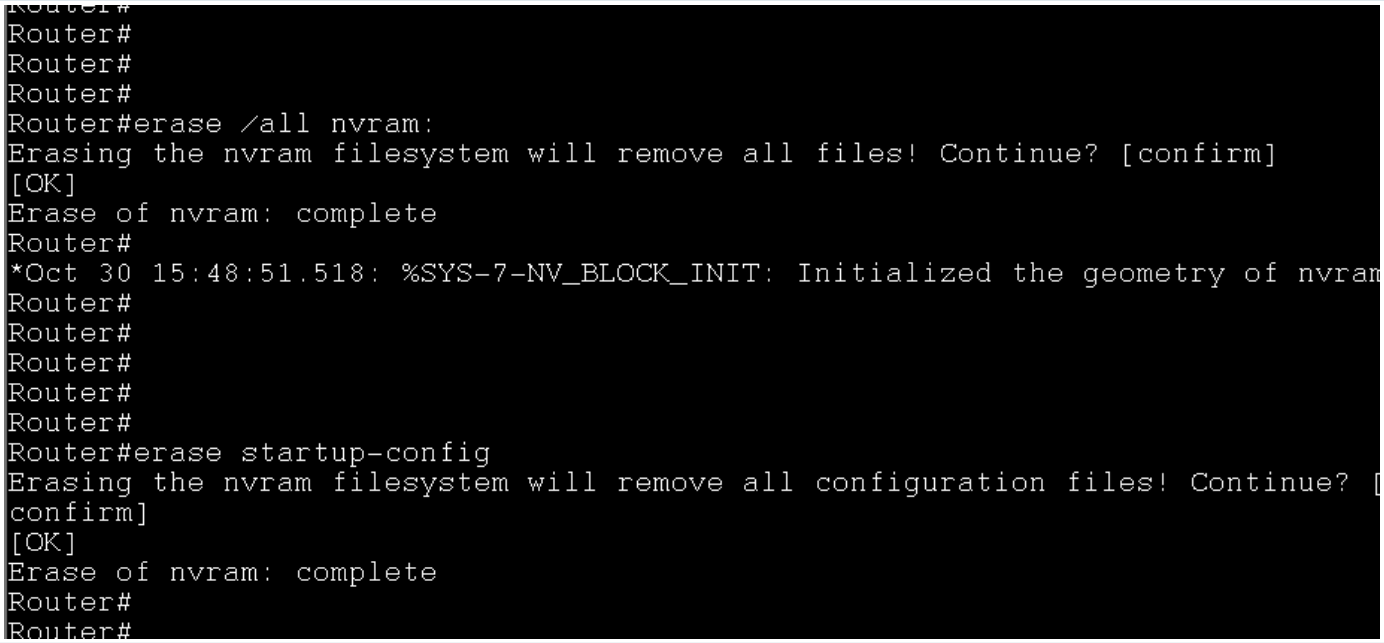
```
Router# erase /all nvram:
Router# erase startup-config
```

WHY THESE COMMANDS WERE USED

erase /all nvram: - Targeted the nonvolatile filesystem for removal.

erase startup-config - Removed the saved startup configuration used during normal boot.

CONSOLE EVIDENCE / REDACTED CAPTURE



STEP 14 / MAINTENANCE PROCEDURE

Bootflash formatting

RECORDED CISCO CLI

```
Router# format ?
Router# format bootflash:
Router# dir bootflash:
```

WHY THESE COMMANDS WERE USED

format ? - Checked available formatting targets.

format bootflash: - Reinitialized the internal flash filesystem.

dir bootflash: - Reviewed the resulting storage listing.

CONSOLE EVIDENCE / REDACTED CAPTURE

```
Router#
Router#
Router#format ?
  bootflash:  Filesystem to be formatted
  usb0:       Filesystem to be formatted

Router#format bootflash:
Format operation may take a while. Continue? [confirm]
Format operation will destroy all data in "bootflash:".  Continue? [confirm]
format completed with no errors

Format of bootflash: complete
Router#
Router#
Router#dir bootflash:
Directory of bootflash:/

   11  drwx            16384  Oct 30 2026 15:52:36 +00:00  lost+found
40801  drwx             4096  Oct 30 2026 15:52:37 +00:00  .prst_sync

2908606464 bytes total (2761994240 bytes free)
Router#
```

STEP 15 / MAINTENANCE PROCEDURE

IOS XE image restoration

RECORDED CISCO CLI

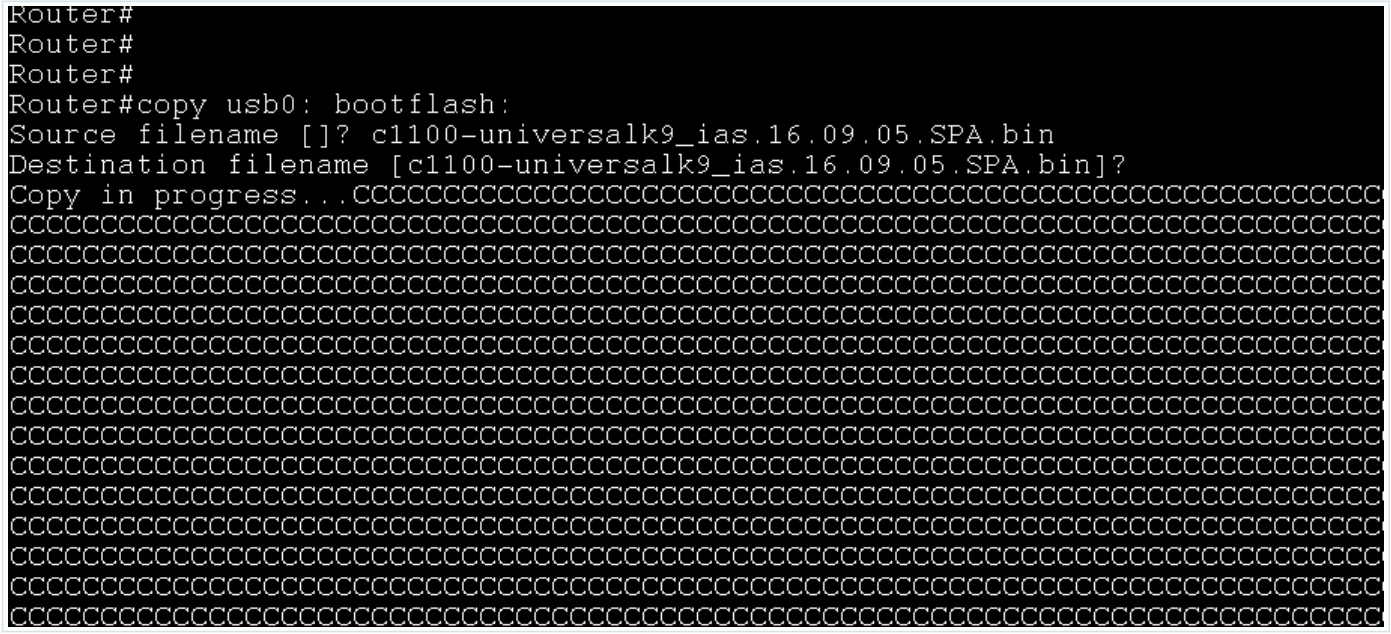
```
Router# copy usb0: bootflash:
Source filename []? c1100-universalk9_ias.16.09.05.SPA.bin
Destination filename [c1100-universalk9_ias.16.09.05.SPA.bin]?
```

WHY THESE COMMANDS WERE USED

copy usb0: bootflash: - Initiated the IOS XE image return from USB to internal flash.

Filename prompts - Selected the image source and destination name.

CONSOLE EVIDENCE / REDACTED CAPTURE



STEP 16 / MAINTENANCE PROCEDURE

Post-operation filesystem review

RECORDED CISCO CLI

```
Router# dir all
```

WHY THESE COMMANDS WERE USED

dir all - Revisited the available filesystems following the recorded storage operations.

CONSOLE EVIDENCE / REDACTED CAPTURE

```
Router#
Router#dir all
Directory of system:/

    2  -r--                0                    <no date>  default-running-c
nfig
    4  dr-x                0                    <no date>  memory

    1  -rw-             1549                    <no date>  running-config
    3  dr-x                0                    <no date>  vfile

No space information available
Directory of tmpsys:/

   73  drw-                0                    <no date>  call-home
    6  drw-                0                    <no date>  eem_lib_system
    5  drw-                0                    <no date>  eem_lib_user
   18  -rw-                0                    <no date>  eem_pnt_0
   20  -rw-                0                    <no date>  eem_pnt_1
   38  -rw-                0                    <no date>  eem_pnt_10
   40  -rw-                0                    <no date>  eem_pnt_11
   42  -rw-                0                    <no date>  eem_pnt_12
   44  -rw-                0                    <no date>  eem_pnt_13
   46  -rw-                0                    <no date>  eem_pnt_14
   48  -rw-                0                    <no date>  eem_pnt_15
   22  -rw-                0                    <no date>  eem_pnt_2
```

STEP 17 / MAINTENANCE PROCEDURE

VLAN inspection

RECORDED CISCO CLI

```
Router# show vlan
```

WHY THESE COMMANDS WERE USED

show vlan - Reviewed VLAN entries present after configuration maintenance.

CONSOLE EVIDENCE / REDACTED CAPTURE

```
Router#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
--
1    default                          active    Gi0/1/0, Gi0/1/1, Gi0/1/2
                                                Gi0/1/3, Gi0/1/4, Gi0/1/5
                                                Gi0/1/6, Gi0/1/7, Wl0/1/8
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Tran
2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ----
-
1    enet  100001     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
-
```

STEP 18 / MAINTENANCE PROCEDURE

Module review and reload

RECORDED CISCO CLI

```
Router# show platform
Router# reload
```

WHY THESE COMMANDS WERE USED

show platform - Rechecked detected modules following maintenance.

reload - Requested a restart using the current boot environment.

CONSOLE EVIDENCE / REDACTED CAPTURE

```
% Invalid input detected at '^' marker.

Router#show platform
Chassis type: C1111-8PWB

Slot      Type                State                 Insert time (ago)
--------- ------------------- --------------------- -----------------
0         C1111-8PWB          ok                    00:10:37
 0/0      C1111-2x1GE         ok                    00:09:02
 0/1      C1111-ES-8          ok                    00:09:02
 0/2      ISR-AP1100AC-B      ok                    00:09:10
R0        C1111-8PWB          ok, active            00:10:37
F0        C1111-8PWB          ok, active            00:10:37
P0        PWR-12V             ok                    00:10:16

Slot      CPLD Version        Firmware Version
--------- ------------------- ----------------------------------------
0         18092702            16.9(1r)
R0        18092702            16.9(1r)
F0        18092702            16.9(1r)

Router#reload
```

STEP 19 / MAINTENANCE PROCEDURE

ROMMON environment review

RECORDED CISCO CLI

```
rommon 1 > set
```

WHY THESE COMMANDS WERE USED

set - Displayed ROMMON environment variables and boot-related settings after the restart attempt.

CONSOLE EVIDENCE / REDACTED CAPTURE

```
rommon 1 >
rommon 1 >
rommon 1 >
rommon 1 > set
PS1=rommon ! >
ETHER_PORT=0
MCP_STARTUP_TRACEFLAGS=00000000:00000000
THRPUT=
EULA_ACCEPTED=TRUE
LICENSE_BOOT_LEVEL=appxk9,appxk9:esg;securityk9,securityk9:esg;
BSI=0
RANDOM_NUM=[REDACTED]
RET_2_RTS=
RET_2_RCALTS=[REDACTED]
rommon 2 >
```

FINAL REVIEW

Results & verification

The report records operations visible in the supplied console evidence. Verified observations are separated from items that require additional evidence to support stronger completion claims.

SUPPORTED BY THE RECORDED EVIDENCE

- Chassis and IOS XE 16.09.05 identified in software and inventory output.
- Interfaces, routes, modules, filesystems and running configuration inspected.
- Selected cryptographic key-maintenance operations recorded.
- NVRAM/startup-config erasure and bootflash formatting operations recorded.
- USB image backup and restoration initiated, with progress captured.
- VLAN, module and ROMMON environment inspected following maintenance.

NOT INDEPENDENTLY VERIFIED IN THE CAPTURES

- Completed backup and restore transfer confirmations or image integrity checks.
- Successful final IOS XE startup after the recorded reload sequence.
- A separate reset or sanitization of the integrated wireless module.
- A formal certification of comprehensive data sanitization.

SOURCE REPOSITORY

View the Cisco ISR 1100 project repository